

Deep Reinforcement Learning

Actor-Critic Algorithms

Prof. Dr. Sebastian Peitz

Chair of Safe Autonomous Systems, TU Dortmund

Summer term 2026

Content

Content

- Two alternative derivations of the policy gradient theorem
 - Bottom-up derivation via the Q-function
 - Sampling with reduced variance
- Advantage functions

Where are we?

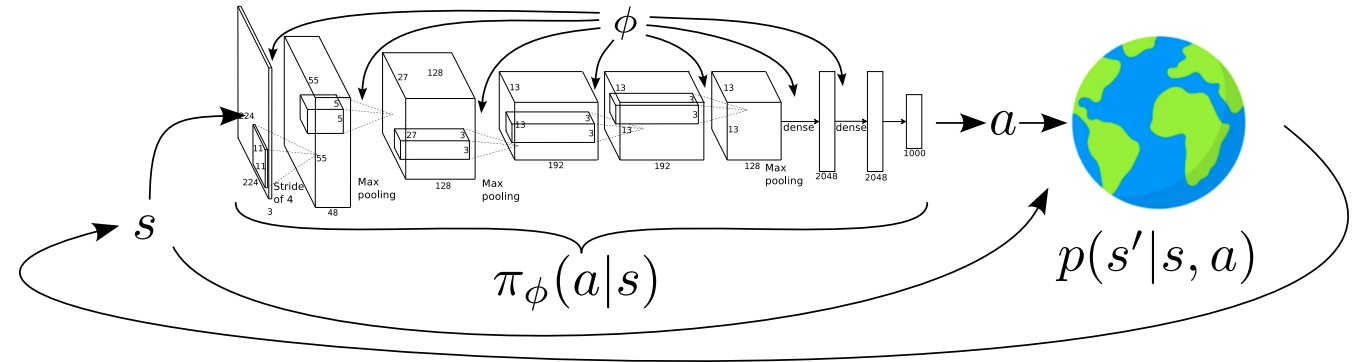
Chapter	Topic	Content
	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6	Brief introduction to deep learning	The basics for what comes next
7	Value function approximation	Value estimation with function approximation
8	Deep Q-learning	Q-learning with neural networks
9	Policy gradients	Direct optimization of the policy
10	Actor-critic algorithms	Improved policy gradients via value functions
11	Advanced algorithms	
	Model-Based Control	
	Advanced Topics	

A more detailed derivation of the policy gradient

The Q-function version of the policy gradient

- Recall that we started with the objective of maximizing the value directly by optimizing over our policy network weights:

$$\begin{aligned}\phi^* &= \arg \max_{\phi} \mathbb{E}_{\tau \sim p_{\phi}(\tau)} \left[\sum_{t=0}^{T-1} r_t \right] \\ &= \arg \max_{\phi} \mathbb{E}_{s_0 \sim p_0} [V_{\phi}(s_0)].\end{aligned}$$



Inspired by Sergey Levine's CS285 lecture.

- Using the definition of the expectation of continuous random variables and the log derivative trick, we arrived at a formulation of the policy gradient from which we can sample using trajectory data ($\tau = (s_0, a_0, \dots, s_{T-1}, a_{T-1}, s_T)$):

$$\nabla_{\phi} L(\phi) = \mathbb{E}_{\tau \sim p_{\phi}(\tau)} \left[\left(\sum_{t=0}^{T-1} \nabla_{\phi} \log \pi_{\phi}(a_t | s_t) \right) \left(\sum_{t=0}^{T-1} r_t \right) \right] \approx \frac{1}{N} \sum_{i=1}^N \left(\sum_{t=0}^{T-1} \nabla_{\phi} \log \pi_{\phi}(a_{i,t} | s_{i,t}) \right) \left(\sum_{t=0}^{T-1} r_{i,t} \right).$$

- **Goal:** derive the policy gradient a second time, but this time using the Q-function instead of the return $\sum_{t=0}^{T-1} r_{i,t}$.

- **Starting point:** the RL objective and its gradient:

$$L(\phi) = \mathbb{E}_{s_0 \sim p_0} [V_\phi(s_0)] \quad \Rightarrow \quad \nabla_\phi L(\phi) = \mathbb{E}_{s_0 \sim p_0} [\nabla_\phi V_\phi(s_0)] = \int_{\mathcal{S}} p_0(s) \nabla_\phi V_\phi(s) \, ds. \quad (1)$$

Deriving the policy gradient (1)

- To derive the policy gradient, we need to take the **gradient of the value function**, i.e., $\nabla_{\phi} V_{\phi}(s)$.
- Our **goal is to introduce the Q-function** (we will see the reason for this later). What's the relation between the Q-function and the value function for continuous actions?

$$\underbrace{V^{\pi}(s) = \sum_{a \in \mathcal{A}} \pi(a|s) Q^{\pi}(s, a)}_{\text{finite } \mathcal{A}} \quad \text{vs.} \quad \underbrace{V^{\pi}(s) = \int_{\mathcal{A}} \pi(a|s) Q^{\pi}(s, a) da}_{\text{continuous } \mathcal{A}}.$$

- Now let's **take the derivative!** Both π_{ϕ} and Q_{ϕ} depend on $\phi \Rightarrow$ product rule ($\frac{d}{dx}(f(x) \cdot g(x)) = f(x) \frac{dg}{dx} + \frac{df}{dx} g(x)$):

$$\begin{aligned} \nabla_{\phi} V_{\phi}(s) &= \nabla_{\phi} \left(\int_{\mathcal{A}} \pi(a|s) Q_{\phi}(s, a) da \right) = \int_{\mathcal{A}} \nabla_{\phi} [\pi(a|s) Q_{\phi}(s, a)] da = \int_{\mathcal{A}} \nabla_{\phi} \pi(a|s) Q_{\phi}(s, a) + \pi(a|s) \nabla_{\phi} Q_{\phi}(s, a) da \\ &= \underbrace{\int_{\mathcal{A}} \nabla_{\phi} \pi(a|s) Q_{\phi}(s, a) da}_{= \xi(s)} + \int_{\mathcal{A}} \pi(a|s) \nabla_{\phi} Q_{\phi}(s, a) da \end{aligned}$$

- In total, we obtain the following **expression** for $\nabla_{\phi} V_{\phi}(s)$:

$$\nabla_{\phi} V_{\phi}(s) = \xi(s) + \int_{\mathcal{A}} \pi(a|s) \nabla_{\phi} Q_{\phi}(s, a) \, da. \quad (2)$$

Deriving the policy gradient (2)

- Before we continue with (2), let's study $Q_\phi(s, a)$ in some more detail, with the goal to find a Bellman recursion formula:

$$Q_\phi(s, a) = r + \int_{\mathcal{S}} p(s'|s, a) V_\phi(s') ds'. \quad (3)$$

- Since neither r nor $p(s'|s, a)$ depend on ϕ , taking the gradient of (3) yields

$$\nabla_\phi Q_\phi(s, a) = \int_{\mathcal{S}} p(s'|s, a) \nabla_\phi V_\phi(s') ds'. \quad (4)$$

- We can substitute this expression into (2) and swap integrals due to linearity:

$$\nabla_\phi V_\phi(s) = \xi(s) + \int_{\mathcal{A}} \pi(a|s) \underbrace{\left[\int_{\mathcal{S}} p(s'|s, a) \nabla_\phi V_\phi(s') ds' \right]}_{(4)} da = \xi(s) + \int_{\mathcal{S}} \underbrace{\left[\int_{\mathcal{A}} \pi(a|s) p(s'|s, a) da \right]}_{=p(s \rightarrow s'|1, \pi_\phi)} \nabla_\phi V_\phi(s') ds'.$$

- Here, $p(s \rightarrow s'|1, \pi_\phi)$ denotes the probability of moving from s to s' in exactly one step, given the policy π_ϕ :

$$\nabla_\phi V_\phi(s) = \xi(s) + \int_{\mathcal{S}} p(s \rightarrow s'|1, \pi_\phi) \nabla_\phi V_\phi(s') ds'. \quad (5)$$

Deriving the policy gradient (3)

- Equation (5) is a Bellman recursion equation, and we can insert the same expression at s' on the right:

$$\nabla_{\phi} V_{\phi}(s) = \xi(s) + \int_{\mathcal{S}} p(s \rightarrow s' | 1, \pi_{\phi}) \underbrace{\left[\xi(s') + \int_{\mathcal{S}} p(s' \rightarrow s'' | 1, \pi_{\phi}) \nabla_{\phi} V_{\phi}(s'') \, ds'' \right]}_{=\nabla_{\phi} V_{\phi}(s')} \, ds'.$$

- We can unroll this expression for an entire trajectory τ of length T . After some resorting, we obtain:

$$\nabla_{\phi} V_{\phi}(s) = \xi(s) + \int_{\mathcal{S}} p(s \rightarrow s' | 1, \pi_{\phi}) \xi(s') \, ds' + \int_{\mathcal{S}} p(s \rightarrow s'' | 2, \pi_{\phi}) \xi(s'') \, ds'' + \dots$$

- We can write this cleanly as a sum over all future timesteps t :

$$\nabla_{\phi} V_{\phi}(s) = \int_{\mathcal{S}} \sum_{t=0}^{T-1} p(s \rightarrow x | t, \pi_{\phi}) \xi(x) \, dx. \quad (6)$$

Note: the previously separate case $\xi(s)$ is included, i.e., $\int_{\mathcal{S}} p(s \rightarrow s | 0, \pi_{\phi}) \xi(x) \, dx = \xi(s)$.

Deriving the policy gradient (4)

- We can now insert (6) into the gradient of our RL objective (Eq. (1)):

$$\nabla_{\phi} L(\phi) = \mathbb{E}_{s_0 \sim p_0} [\nabla_{\phi} V_{\phi}(s_0)] = \int_{\mathcal{S}} p_0(s) \nabla_{\phi} V_{\phi}(s) \, ds = \int_{\mathcal{S}} p_0(s) \left[\int_{\mathcal{S}} \sum_{t=0}^{T-1} p(s \rightarrow x|t, \pi_{\phi}) \xi(x) \, dx \right] ds.$$

- By changing the order of integration, we isolate the total expected time spent in state x :

$$\nabla_{\phi} L(\phi) = \int_{\mathcal{S}} \left[\int_{\mathcal{S}} p_0(s) \sum_{t=0}^{T-1} p(s \rightarrow x|t, \pi_{\phi}) \, ds \right] \xi(x) \, dx.$$

- The term inside the brackets is the **undiscounted state visitation frequency** (or the expected number of times state x is visited in an episode), which we denote as $\eta_{\phi}(s)$:

$$\eta_{\phi}(s) = \sum_{t=0}^{T-1} p(s_t = s | \pi_{\phi}). \tag{7}$$

- This simplifies our expression to:

$$\nabla_{\phi} L(\phi) = \int_{\mathcal{S}} \eta_{\phi}(s) \xi(s) \, ds.$$

Deriving the policy gradient (5)

- Reintroducing the full definition of $\xi(s) = \int_{\mathcal{A}} \nabla_{\phi} \pi(a|s) Q_{\phi}(s, a) da$ (and keeping s as our state variable):

$$\nabla_{\phi} L(\phi) = \int_{\mathcal{S}} \eta_{\phi}(s) \int_{\mathcal{A}} \nabla_{\phi} \pi_{\phi}(a|s) Q_{\phi}(s, a) da ds.$$

- Apply the log-derivative identity:

$$\nabla_{\phi} L(\phi) = \int_{\mathcal{S}} \eta_{\phi}(s) \int_{\mathcal{A}} \pi_{\phi}(a|s) \nabla_{\phi} \log \pi_{\phi}(a|s) Q_{\phi}(s, a) da ds.$$

- In conclusion, the gradient can be expressed using the unnormalized state visitation measure $\eta_{\phi}(s)$:

$$\nabla_{\phi} L(\phi) = \mathbb{E}_{\tau \sim p_{\phi}(\tau)} \left[\sum_{t=0}^{T-1} \nabla_{\phi} \log \pi_{\phi}(a_t|s_t) Q_{\phi}(s_t, a_t) \right]. \quad (8)$$

Expectations in continuous spaces

The expected value of a function $f(s)$ is

$$\mathbb{E}_{s \sim p} [f(s)] = \int p(s) f(s) ds.$$

Here, p is the density according to which s is distributed, with $\int p(s) ds = 1$.

A convenient identity

$$\nabla_{\phi} \pi_{\phi}(a|s) = \pi_{\phi}(a|s) \nabla_{\phi} \log \pi_{\phi}(a|s)$$

Note: In the infinite-horizon case, (7) becomes $\eta_\phi(s) = \sum_{t=0}^{\infty} p(s_t = s | p_0, \pi_\phi) \Rightarrow$ Equation (8) becomes

$$\nabla_\phi L(\phi) = \mathbb{E}_{s \sim \eta_\phi, a \sim \pi_\phi} [\nabla_\phi \log \pi_\phi(a|s) Q_\phi(s, a)].$$

Comparison of the two formulations

Formulation via reward trajectories (**sampling version in blue**),

$$\nabla_{\phi} L(\phi) = \mathbb{E}_{\tau \sim p_{\phi}(\tau)} \left[\sum_{t=0}^{T-1} \nabla_{\phi} \log \pi_{\phi}(a_t | s_t) \left(\sum_{t'=t}^{T-1} r_{t'} \right) \right] \approx \frac{1}{N} \sum_{i=1}^N \left(\sum_{t'=t}^{T-1} \nabla_{\phi} \log \pi_{\phi}(a_{i,t} | s_{i,t}) \left(\sum_{t'=t}^{T-1} r_{i,t'} \right) \right).$$

Derivation via formal relation between V and Q (Eq. (8)),

$$\nabla_{\phi} L(\phi) = \mathbb{E}_{\tau \sim p_{\phi}(\tau)} \left[\sum_{t=0}^{T-1} \nabla_{\phi} \log \pi_{\phi}(a_t | s_t) Q_{\phi}(s_t, a_t) \right] \approx \frac{1}{N} \sum_{i=1}^N \left(\sum_{t=0}^{T-1} \nabla_{\phi} \log \pi_{\phi}(a_{i,t} | s_{i,t}) Q_{\phi}(s_{i,t}, a_{i,t}) \right).$$

A much simpler way to arrive at this formulation

- Observe that the **reward-to-go** $\hat{Q}_{i,t} = \sum_{t'=t}^{T-1} r_{i,t'}$ is an **unbiased estimate** of $Q(s_t, a_t)$.
- Also, $\hat{Q}_{i,t}$ is a single-sample estimate, and summing over N samples gives us a **batch estimate**.

- To reduce the variance, it would be a lot better to simply use the **true expected reward-to-go**:

$$\sum_{t'=t}^{T-1} \mathbb{E}_{\pi_{\phi}} [r_{t'} | s_t, a_t] = Q_{\phi}(s_{i,t}, a_{i,t}).$$

References